Number Theory

Independent diagnostic, two proof questions

Time allowed: 45 minutes **Total: 14 marks**

Candidate identifier: _____ **Date:** _____

Instructions. Attempt both questions independently. Show complete arguments, including all relevant cases and any final verification. No internet, calculator, AI tools or hints during the timed sitting. If unfinished, write down any useful progress. Stop the timed work when time is called.

Question 1 **[7 marks]**

Prove that 30 divides $n^5 - n$ for every integer n. Your proof should explain the divisibility by each of the primes 2, 3 and 5.

Question 2 **[7 marks]**

For each positive integer n, determine

$$\gcd\bigl(n^2 + 1, (n+1)^2 + 1\bigr).$$

Give an exact answer depending on n, and prove that no other values of this greatest common divisor are possible.

After the sitting (not part of the timed assessment): Record approximate time spent on each question, whether either problem was familiar, and whether you worked beyond the limit. If you need more space, use the separate answer sheets.

Q1: _____ min Q2: _____ min Prior exposure: _____